

Understanding Boosting Through Margins

Exponential vs Logistic Loss

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Binary Classification Setup

Additive model:

$$F(x) = \sum_{m=1}^M \alpha_m h_m(x)$$

Prediction rule:

$$\hat{y} = \text{sign}(F(x))$$

If $y \in \{-1, +1\}$,

Correct classification:

$$yF(x) > 0$$

This quantity $yF(x)$ is called the **margin**.

Why Margin?

Define:

$$\gamma(x) = yF(x)$$

- $\gamma > 0$ correct
- $\gamma < 0$ wrong
- $|\gamma|$ measures confidence

Boosting increases margins.

From 0–1 Loss to Surrogate Loss

True error:

$$\mathbb{I}(yF < 0)$$

Hard to optimize.

Upper bound:

$$\mathbb{I}(yF < 0) \leq e^{-yF}$$

Hence exponential loss:

$$L = e^{-yF}$$

Logistic loss:

$$L = \log(1 + e^{-yF})$$

Both depend only on margin.

Dataset

x	y
1	-1
2	-1
3	+1
4	-1
5	+1
6	+1

Single stump cannot separate perfectly.

How Are Stumps Chosen?

Candidate splits:

1.5, 2.5, 3.5, 4.5, 5.5

For exponential: Minimize weighted classification error.

For logistic: Minimize squared residual error:

$$\sum (r_i - c_{region})^2$$

Iteration 1 — Choose Best Split

Best split at 3.5

Error:

$$\epsilon_1 = \frac{2}{6} = 0.333$$

$$\alpha_1 = \frac{1}{2} \ln \frac{1 - \epsilon}{\epsilon} = 0.3466$$

$$F^{(1)} = \alpha_1 h_1$$

Margins After Iteration 1

x	$yF^{(1)}$
1	0.3466
2	0.3466
3	-0.3466
4	-0.3466
5	0.3466
6	0.3466

Two negative margins.

Iteration 2

Reweight.

Best split at 4.5.

$$\alpha_2 = 0.255$$

$$F^{(2)} = F^{(1)} + \alpha_2 h_2$$

Margins After Iteration 2

x	$yF^{(2)}$
1	0.6016
2	0.6016
3	-0.0916
4	0.0916
5	0.0916
6	0.0916

Only one negative margin remains.

Iteration 3

$$\alpha_3 = 0.15$$

All margins positive:

x	$yF^{(3)}$
1	0.7516
2	0.7516
3	0.0584
4	0.0584
5	0.2416
6	0.2416

Training error = 0.

Logistic Gradient Derivation

Loss:

$$L = \log(1 + e^{-yF})$$

Derivative:

$$\frac{\partial L}{\partial F} = -\frac{y}{1 + e^{yF}}$$

Pseudo-residual:

$$r_i = \frac{y_i}{1 + e^{y_i F}}$$

Iteration 1 — Logistic

Initialize:

$$F^{(0)} = 0$$

Residuals:

$$r_i^{(1)} = \frac{y_i}{2}$$

x	$r^{(1)}$
1	-0.5
2	-0.5
3	0.5
4	-0.5
5	0.5
6	0.5

Choose Split via SSE

Split at 3.5:

Left mean:

$$(-0.5 - 0.5 + 0.5)/3 = -0.167$$

Right mean:

$$(-0.5 + 0.5 + 0.5)/3 = 0.167$$

Compute total SSE:

$$\sum (r_i - \bar{r}_{region})^2$$

This split minimizes SSE among candidates.

Iteration 2 — Logistic Residuals

Update:

$$F^{(1)} = h_1$$

Compute new residuals:

Example:

$$r_1 = \frac{-1}{1 + e^{0.167}} = -0.458$$

Full table:

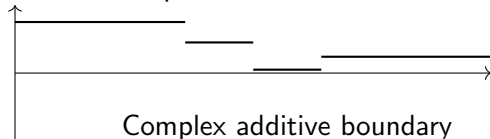
x	$r^{(2)}$
1	-0.458
2	-0.458
3	0.542
4	-0.542
5	0.458
6	0.458

Decision Boundary Evolution

After 1 stump: single threshold at 3.5

After 2 stumps: thresholds at 3.5 and 4.5

After 3 stumps:



Boosting builds piecewise function.

Boosting Advantage

Single stump: 66.7% accuracy

Boosted ensemble: 100%

Margin increases Loss decreases Boundary becomes flexible

Boosting performs:

Functional Gradient Descent

Sequentially correcting errors.